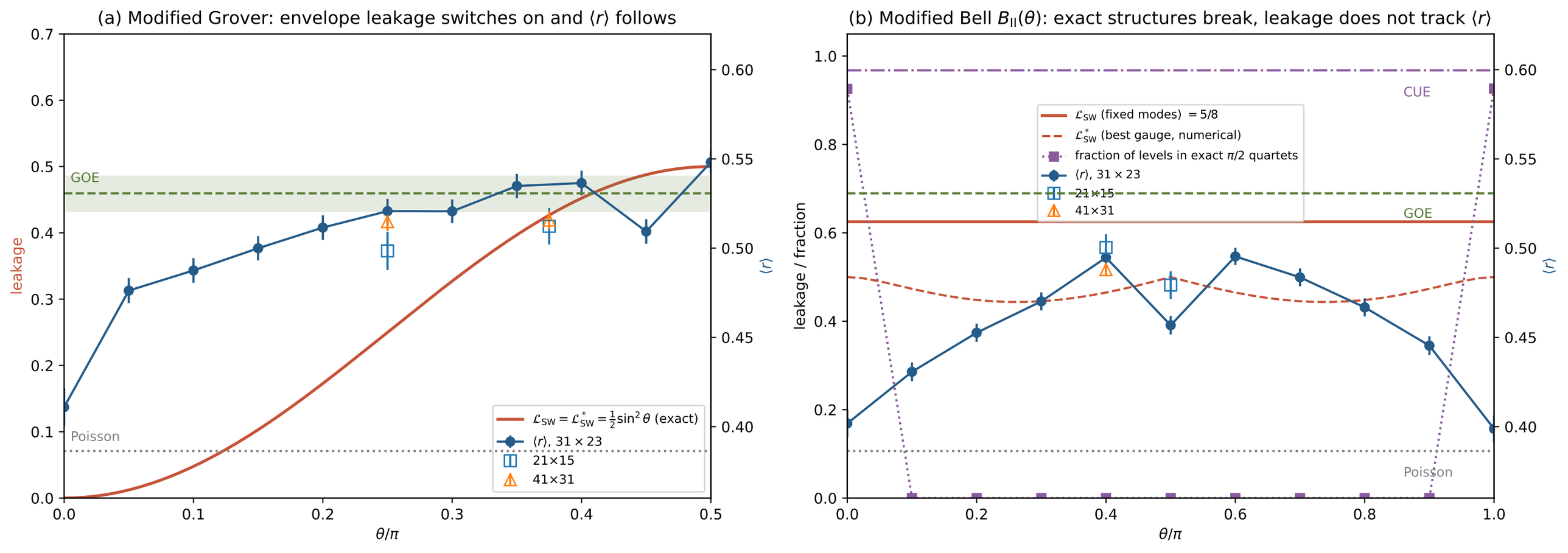
(a) Modified Grover: envelope leakage switches on and ⟨r⟩ follows
leakage
⟨r⟩
θ/π
GOE
Poisson
$\mathcal{L}_{SW} = \mathcal{L}^*_{SW} = \frac{1}{2}\sin^2\theta$ (exact)
⟨r⟩, 31 × 23
21×15
41×31
(b) Modified Bell $B_{II}(\theta)$: exact structures break, leakage does not track ⟨r⟩
leakage / fraction
⟨r⟩
θ/π
CUE
GOE
Poisson
$\mathcal{L}_{SW}$ (fixed modes) = 5/8
$\mathcal{L}^*_{SW}$ (best gauge, numerical)
fraction of levels in exact π/2 quartets
⟨r⟩, 31 × 23
21×15
41×31